화학수학

다섯 번째 수업

다변수 미분

다변수 적분

Pfaffian Form: General Differential Form

(파피안형)

$$\mathrm{d}u = M(x,y)\mathrm{d}x + N(x,y)\mathrm{d}y$$

- Exact differential (완전 미분):

$$M(x,y) = \left(\frac{\partial u}{\partial x}\right)_y, \qquad N(x,y) = \left(\frac{\partial u}{\partial y}\right)_x$$

- Inexact differential (불완전 미분): otherwise

Test of Exactness

$$\mathrm{d}u = M(x, y)\mathrm{d}x + N(x, y)\mathrm{d}y$$

Using the Euler reciprocity relation,

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$

if the differential is exact,

$$\left(\frac{\partial N}{\partial x}\right)_y = \left(\frac{\partial M}{\partial y}\right)_x$$

Integrating Factor

(적분인자)

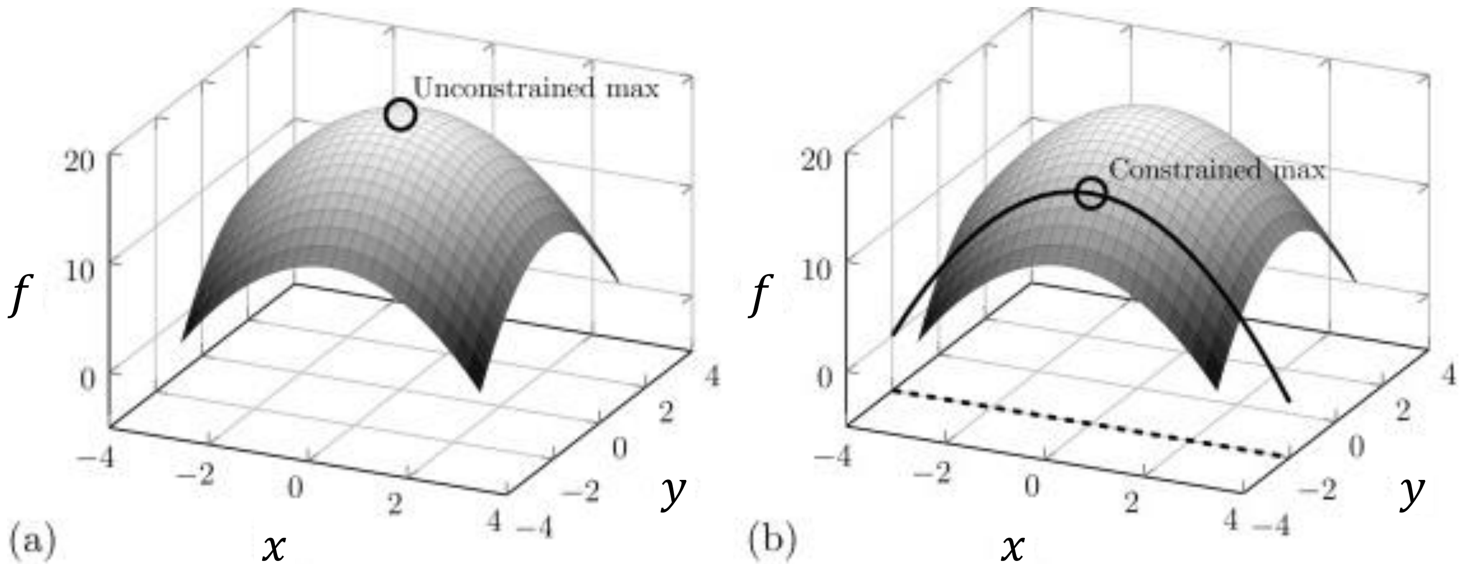
inexact differential → exact differential

by multiplying a function (integrating factor)

No general method to find an integrating factor

Constrained Optimization

(제약 최적화)



with a constraint (제약 조건)

(Image: Garrett, *Introduction to Actuarial and Financial Mathematical Methods* (2015), Figure 12.6)

Lagrange's Method

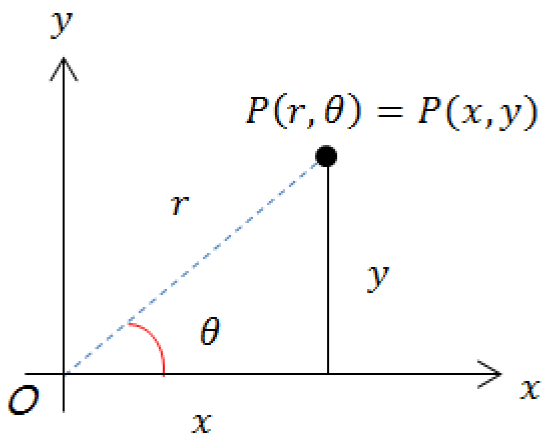
Find the constrained extremum in $f(x, y)$,

when the constraint is written in the form $g(x, y) = 0$

1. New function $u(x, y) = f(x, y) + \lambda g(x, y)$
2. Partial derivatives of u = 0
3. Solve the equations from point 2 and $g(x, y) = 0$ to obtain x, y, and λ → constrained extrema

Coordinate Systems

- 2 dimensions
  - Cartesian coordinates (직교 좌표)
  - Polar coordinates (극좌표)

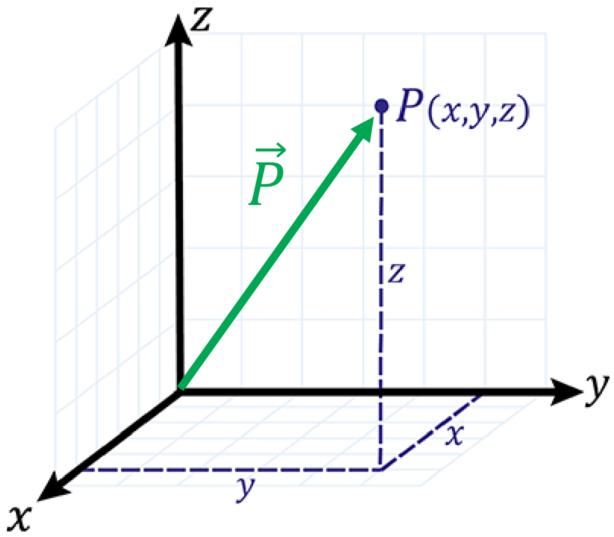


- 3 dimensions
  - Cartesian coordinates (직교 좌표)
  - Spherical (polar) coordinates (구면 좌표)
  - Cylindrical (polar) coordinates (원통 좌표)

(Image: Fadl and Semary, *Neurocomputing* (2017), Figure 3)

Vector

A quantity with magnitude and direction



(Image: https://www.skillsyouneed.com/num/cartesian-coordinates.html)

Scalar Product (내적)

The scalar product (dot product) of $\vec{A}$ and $\vec{B}$ is

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

and, equivalently,

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos \alpha$$

where α is the angle between the two vectors.

Vector Product (외적)

The vector product (cross product) of $\vec{A}$ and $\vec{B}$ is

$$\vec{A} \times \vec{B}$$

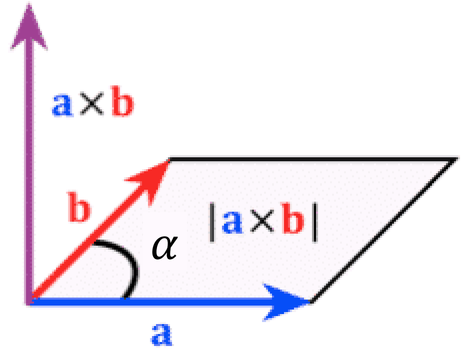
$$= \hat{\imath}\left(A_y B_z - A_z B_y\right) + \hat{\jmath}(A_z B_x - A_x B_z) + \hat{k}\left(A_x B_y - A_y B_x\right)$$

$$= \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Geometric Definition of Vector Product

For $\vec{C} = \vec{A} \times \vec{B}$,

- Magnitude: $|\vec{C}| = |\vec{A}||\vec{B}| \sin\alpha$ (α: angle between $\vec{A}$ & $\vec{B}$)
- Direction: right-hand rule



(Image: https://physicsinmyview.com/2020/10/dot-product-vs-cross-product.html)

오늘의 수업

- Vector derivatives
  - Gradient, divergence, curl, Laplacian
  - In different coordinate systems
- Line integral
  - Definition
  - Exact and inexact differentials
  - Applications: thermodynamics

How to Differentiate Vectors?

- Scalar by scalar: $\mathrm{d}f/\mathrm{d}x$ → We know how to
- Scalar by vector: $\mathrm{d}f/\mathrm{d}\vec{r}$

→ "Gradient"

- Vector by scalar: $\mathrm{d}\vec{F}/\mathrm{d}x$

→ Straightforward: $\mathrm{d}\vec{F}/\mathrm{d}x = \left(\mathrm{d}F_x/\mathrm{d}x\,, \mathrm{d}F_y/\mathrm{d}x\,, \mathrm{d}F_z/\mathrm{d}x\right)$

- Vector by vector: $\mathrm{d}\vec{F}/\mathrm{d}\vec{r}$

→ "Divergence" and "curl"

Vector Derivatives

- Gradient (기울기 연산자) ∇f

  scalar → vector

- Divergence (발산 연산자) $\nabla \cdot \vec{F}$

  vector → scalar

- Curl (회전 연산자) $\nabla \times \vec{F}$

  vector → vector

- Laplacian (라플라스 연산자) $\nabla^2 f$

  scalar → scalar (2nd order)

Gradient

$$\nabla f = \hat{\imath}\left(\frac{\partial f}{\partial x}\right) + \hat{\jmath}\left(\frac{\partial f}{\partial y}\right) + \hat{k}\left(\frac{\partial f}{\partial z}\right) = \mathrm{grad}\ f$$

($\hat{\imath}, \hat{\jmath}, \hat{k}$: unit vectors)

Example 8.18

Find the gradient of the function $f(x, y, z) = x^2 + y^2 + z^2$.

Using the definition of the gradient,

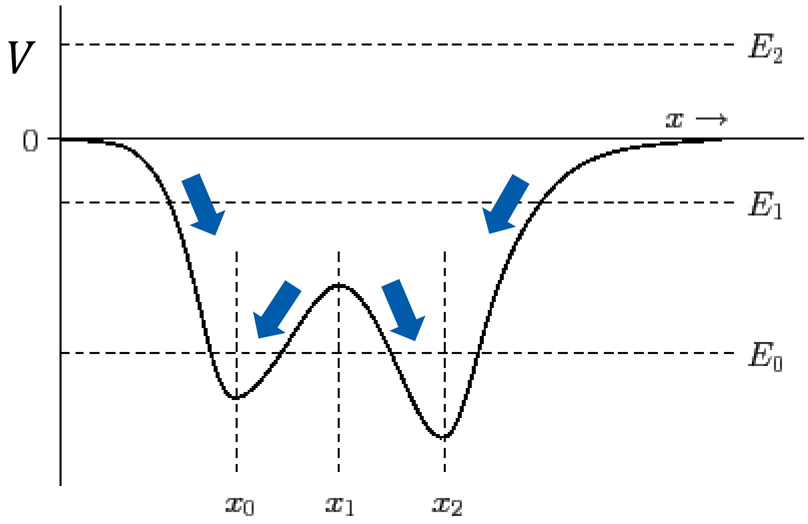
$$\nabla f = \hat{\imath}\left(\frac{\partial f}{\partial x}\right) + \hat{\jmath}\left(\frac{\partial f}{\partial y}\right) + \hat{k}\left(\frac{\partial f}{\partial z}\right)$$

$$\nabla f = \hat{\imath}(2x) + \hat{\jmath}(2y) + \hat{k}(2z) = 2(x\hat{\imath} + y\hat{\jmath} + z\hat{k})$$

Example: Force and Potential

$$\vec{F} = -\nabla V$$

where $\vec{F}$ is the force and V is the potential energy



(Image: http://farside.ph.utexas.edu/teaching/336k/Newtonhtml/node16.html)

Example 8.19

Neglecting the attractions of all other celestial bodies, the gravitational potential energy of the earth and the sun is given by

$$V = -\frac{Gm_s m_e}{r},$$

where G is the universal gravitational constant, m_s and m_e are the masses of the sun and the earth, respectively, and r is the distance between the centers of the sun and the earth. Express the force on the earth.

Using the previous formula,

$$\vec{F} = -\nabla V = \nabla\left(\frac{Gm_s m_e}{r}\right) = Gm_s m_e \nabla\left(\frac{1}{r}\right) = Gm_s m_e \nabla\left(\frac{1}{\sqrt{x^2 + y^2 + z^2}}\right)$$

Example 8.19

$$\vec{F} = Gm_s m_e \nabla\left(\frac{1}{\sqrt{x^2+y^2+z^2}}\right)$$

$$\frac{\partial}{\partial x}\left(\frac{1}{\sqrt{x^2+y^2+z^2}}\right) = -\frac{1}{2}\frac{2x}{(x^2+y^2+z^2)^{3/2}} = -\frac{x}{r^3}$$

Similarly,

$$\frac{\partial}{\partial y}\left(\frac{1}{\sqrt{x^2+y^2+z^2}}\right) = -\frac{y}{r^3}, \qquad \frac{\partial}{\partial z}\left(\frac{1}{\sqrt{x^2+y^2+z^2}}\right) = -\frac{z}{r^3}$$

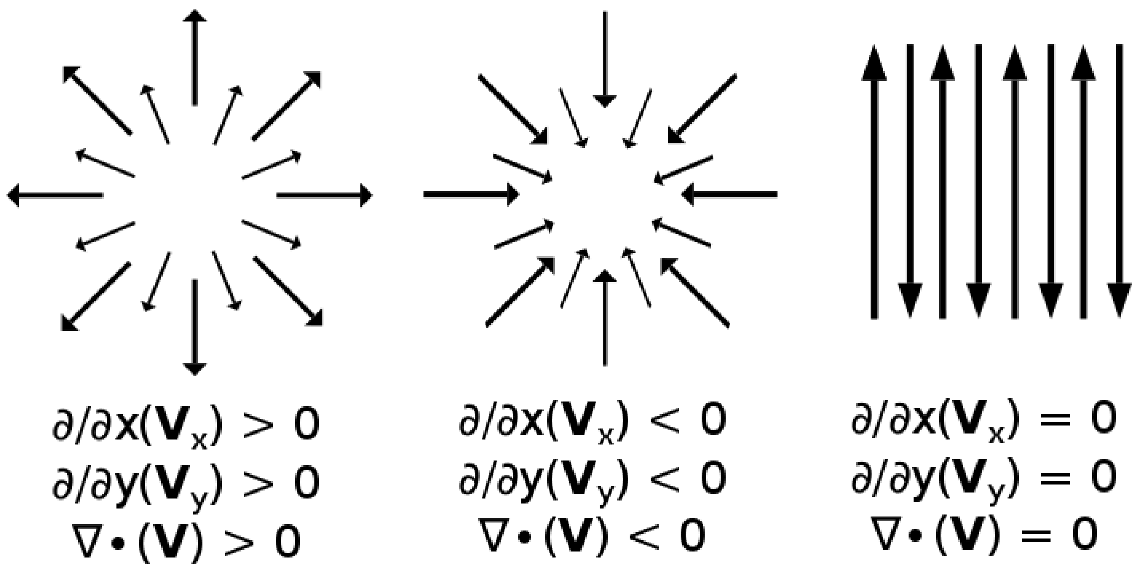
Hence,

$$\vec{F} = Gm_s m_e\left(-\frac{x}{r^3}\hat{\imath} - \frac{y}{r^3}\hat{\jmath} - \frac{z}{r^3}\hat{k}\right) = -\frac{Gm_s m_e}{r^3}\left(x\hat{\imath} + y\hat{\jmath} + z\hat{k}\right)$$

Divergence

$$\nabla \cdot \vec{F} = \left(\frac{\partial F_x}{\partial x}\right) + \left(\frac{\partial F_y}{\partial y}\right) + \left(\frac{\partial F_z}{\partial z}\right) = \mathrm{div}\,\vec{F}$$

Meaning of Divergence



(Image: https://commons.wikimedia.org/wiki/File:Divergence_(captions).svg)

Example 8.20

Find $\nabla \cdot \vec{F}$ if $\vec{F}$ is given by

$$\vec{F} = \hat{\imath}x^2 + \hat{\jmath}yz + \hat{k}\frac{xz^2}{y}.$$

Using the definition of the divergence,

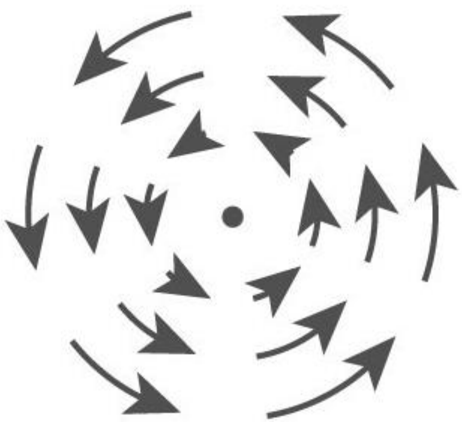
$$\nabla \cdot \vec{F} = \left(\frac{\partial F_x}{\partial x}\right) + \left(\frac{\partial F_y}{\partial y}\right) + \left(\frac{\partial F_z}{\partial z}\right)$$

$$\nabla \cdot \vec{F} = 2x + z + \frac{2xz}{y}$$

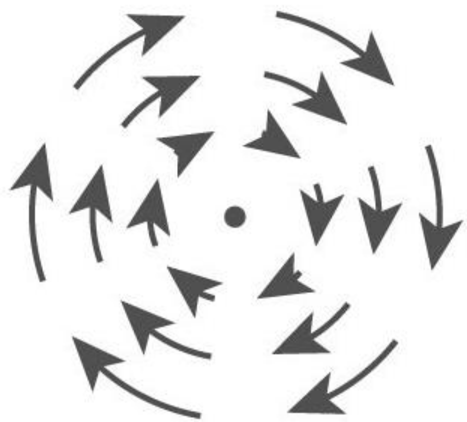
Curl

$$\nabla \times \vec{F} = \hat{\imath}\left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right) + \hat{\jmath}\left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right) + \hat{k}\left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)$$
$$= \text{curl } \vec{F}$$

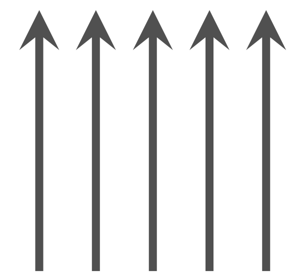
Meaning of Curl



$$\left(\nabla \times \vec{F}\right)_z > 0$$



$$\left(\nabla \times \vec{F}\right)_z < 0$$



$$\left(\nabla \times \vec{F}\right)_z = 0$$

(Image: https://www.youtube.com/watch?v=OK4w44GHJNc)

Example 8.21

Find $\nabla \times \vec{F}$ if $\vec{F}$ is $\vec{F} = \hat{\imath}y + \hat{\jmath}z + \hat{k}x$.

Using the definition of the curl,

$$\nabla \times \vec{F} = \hat{\imath}\left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right) + \hat{\jmath}\left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right) + \hat{k}\left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)$$

$$\nabla \times \vec{F} = \hat{\imath}(0-1) + \hat{\jmath}(0-1) + \hat{k}(0-1) = -\hat{\imath} - \hat{\jmath} - \hat{k}$$

Laplacian: divergence of gradient

$$\nabla^2 f = \left(\frac{\partial^2 f}{\partial x^2}\right) + \left(\frac{\partial^2 f}{\partial y^2}\right) + \left(\frac{\partial^2 f}{\partial z^2}\right)$$

Example 8.22

Find the Laplacian of the function

$$f(x, y, z) = A\sin(ax)\sin(by)\sin(cz).$$

Using the definition of the Laplacian,

$$\nabla^2 f = \left(\frac{\partial^2 f}{\partial x^2}\right) + \left(\frac{\partial^2 f}{\partial y^2}\right) + \left(\frac{\partial^2 f}{\partial z^2}\right)$$

$$(\partial^2 f/\partial x^2) = -Aa^2\sin(ax)\sin(by)\sin(cz) = -a^2 f(x, y, z)$$

$$(\partial^2 f/\partial y^2) = -Ab^2\sin(ax)\sin(by)\sin(cz) = -b^2 f(x, y, z)$$

$$(\partial^2 f/\partial z^2) = -Ac^2\sin(ax)\sin(by)\sin(cz) = -c^2 f(x, y, z)$$

Hence,

$$\nabla^2 f = -(a^2 + b^2 + c^2) f(x, y, z)$$

Infinitesimal Displacements

The infinitesimal displacement

$$\vec{r} \to \vec{r} + \mathrm{d}\vec{r}$$

can be represented as

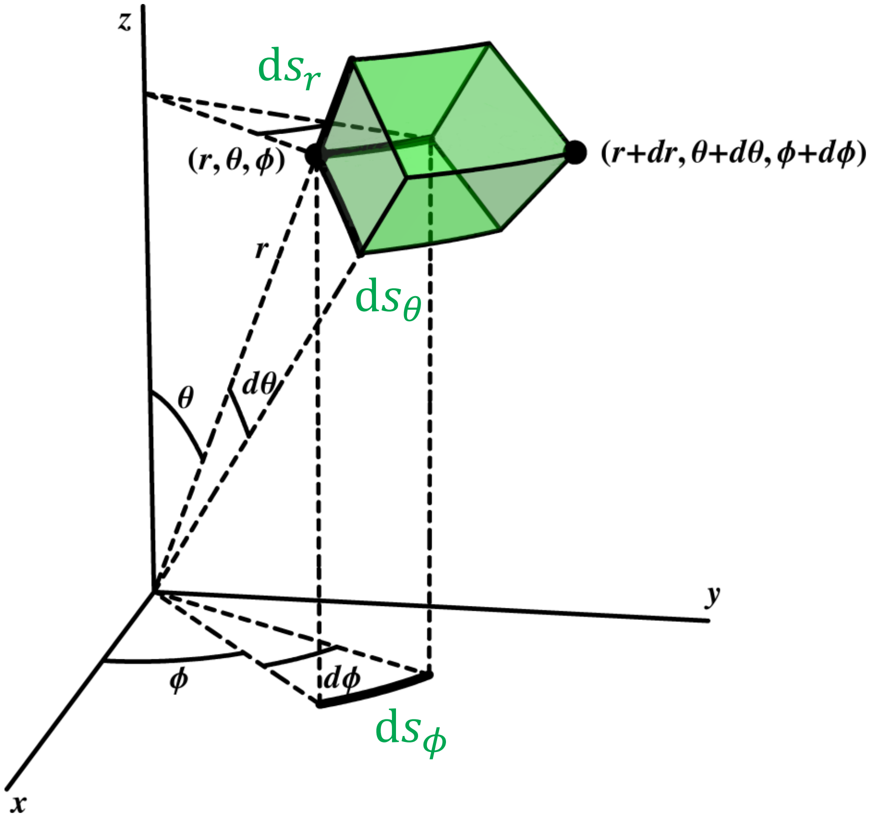
1. in Cartesian coordinates

$$\mathrm{d}\vec{r} = \hat{\imath}\mathrm{d}x + \hat{\jmath}\mathrm{d}y + \hat{k}\mathrm{d}z$$

2. in spherical polar coordinates

$$\mathrm{d}\vec{r} = \hat{e}_r \mathrm{d}s_r + \hat{e}_\theta \mathrm{d}s_\theta + \hat{e}_\phi \mathrm{d}s_\phi$$

In the Spherical Coordinates



$$ds_r = dr$$

$$ds_\theta = r\,d\theta$$

$$ds_\phi = r\sin\theta\,d\phi$$

(Image: https://bridge.math.oregonstate.edu/Book/dscoord.html)

Infinitesimal Displacements

In Cartesian coordinates,

$$\mathrm{d}\vec{r} = \hat{\imath}\,\mathrm{d}x + \hat{\jmath}\,\mathrm{d}y + \hat{k}\,\mathrm{d}z$$

In spherical polar coordinates,

$$\mathrm{d}\vec{r} = \hat{e}_r \mathrm{d}r + \hat{e}_\theta r\,\mathrm{d}\theta + \hat{e}_\phi r \sin\theta\,\mathrm{d}\phi$$

In general,

$$\mathrm{d}\vec{r} = \hat{e}_1 h_1 \mathrm{d}q_1 + \hat{e}_2 h_2 \mathrm{d}q_2 + \hat{e}_3 h_3 \mathrm{d}q_3$$

Gradient

$$\nabla f = \hat{\imath}\frac{\partial f}{\partial x} + \hat{\jmath}\frac{\partial f}{\partial y} + \hat{k}\frac{\partial f}{\partial z}$$

In the other coordinate system,

$$\nabla f = \hat{e}_1 \frac{1}{h_1}\frac{\partial f}{\partial q_1} + \hat{e}_2 \frac{1}{h_2}\frac{\partial f}{\partial q_2} + \hat{e}_3 \frac{1}{h_3}\frac{\partial f}{\partial q_3}$$

In spherical polar coordinates,

$$\nabla f = \hat{e}_r \frac{\partial f}{\partial r} + \hat{e}_\theta \frac{1}{r}\frac{\partial f}{\partial \theta} + \hat{e}_\phi \frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}$$

Divergence

$$\nabla \cdot \vec{F} = \left(\frac{\partial F_x}{\partial x}\right) + \left(\frac{\partial F_y}{\partial y}\right) + \left(\frac{\partial F_z}{\partial z}\right)$$

For $\vec{F} = \hat{e}_1 F_1 + \hat{e}_2 F_2 + \hat{e}_3 F_3$,

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial(F_1 h_2 h_3)}{\partial q_1} + \frac{\partial(F_2 h_1 h_3)}{\partial q_2} + \frac{\partial(F_3 h_1 h_2)}{\partial q_3}\right]$$

In spherical polar coordinates,

$$\nabla \cdot \vec{F} = \frac{1}{r^2 \sin\theta}\left[\frac{\partial(F_r r^2 \sin\theta)}{\partial r} + \frac{\partial(F_\theta r \sin\theta)}{\partial \theta} + \frac{\partial\left(F_\phi r\right)}{\partial \phi}\right]$$

Example 8.23(b)

Find the divergence of the position vector, which in spherical polar coordinates is $\vec{r} = \hat{e}_r r$.

Using the previous formula,

$$\nabla \cdot \vec{F} = \frac{1}{r^2 \sin\theta}\left[\frac{\partial(F_r r^2 \sin\theta)}{\partial r} + \frac{\partial(F_\theta r \sin\theta)}{\partial\theta} + \frac{\partial\left(F_\phi r\right)}{\partial\phi}\right]$$

Since $F_r = r, F_\theta = 0, F_\phi = 0$,

$$\nabla \cdot \vec{F} = \frac{1}{r^2 \sin\theta}\frac{\partial(r^3 \sin\theta)}{\partial r} = \frac{3r^2 \sin\theta}{r^2 \sin\theta} = 3$$

Curl

For $\vec{F} = \hat{e}_1 F_1 + \hat{e}_2 F_2 + \hat{e}_3 F_3$,

$$\nabla \times \vec{F} = \hat{e}_1 \frac{1}{h_2 h_3}\left[\frac{\partial(h_3 F_3)}{\partial q_2} - \frac{\partial(h_2 F_2)}{\partial q_3}\right] + \hat{e}_2 \frac{1}{h_1 h_3}\left[\frac{\partial(h_1 F_1)}{\partial q_3} - \frac{\partial(h_3 F_3)}{\partial q_1}\right] + \hat{e}_3 \frac{1}{h_1 h_2}\left[\frac{\partial(h_2 F_2)}{\partial q_1} - \frac{\partial(h_1 F_1)}{\partial q_2}\right]$$

Curl in Spherical Coordinates

For $\vec{F} = \hat{e}_r F_r + \hat{e}_\theta F_\theta + \hat{e}_\phi F_\phi$,

$$\nabla \times \vec{F} = \hat{e}_r \frac{1}{r^2 \sin\theta}\left[\frac{\partial\left(r\sin\theta\, F_\phi\right)}{\partial\theta} - \frac{\partial(rF_\theta)}{\partial\phi}\right]$$

$$+ \hat{e}_\theta \frac{1}{r\sin\theta}\left[\frac{\partial F_r}{\partial\phi} - \frac{\partial\left(r\sin\theta\, F_\phi\right)}{\partial r}\right]$$

$$+ \hat{e}_\phi \frac{1}{r}\left[\frac{\partial(rF_\theta)}{\partial r} - \frac{\partial F_r}{\partial\theta}\right]$$

Example 8.24

Find the curl of the function $\vec{F} = \hat{e}_r \sin\phi$.

Using the previous formula,

$$\nabla \times \vec{F} = \hat{e}_r \frac{1}{r^2 \sin\theta}\left[\frac{\partial\left(r\sin\theta\, F_\phi\right)}{\partial\theta} - \frac{\partial(rF_\theta)}{\partial\phi}\right] + \hat{e}_\theta \frac{1}{r\sin\theta}\left[\frac{\partial F_r}{\partial\phi} - \frac{\partial\left(r\sin\theta\, F_\phi\right)}{\partial r}\right]$$
$$+ \hat{e}_\phi \frac{1}{r}\left[\frac{\partial(rF_\theta)}{\partial r} - \frac{\partial F_r}{\partial\theta}\right]$$

Since $F_r = \sin\phi\,, F_\theta = 0, F_\phi = 0$,

$$\nabla \times \vec{F} = \hat{e}_\theta \frac{1}{r\sin\theta}\frac{\partial(\sin\phi)}{\partial\phi} + \hat{e}_\phi \frac{1}{r}\left[-\frac{\partial(\sin\phi)}{\partial\theta}\right] = \hat{e}_\theta \frac{\cos\phi}{r\sin\theta}$$

Laplacian

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial q_1}\left(\frac{h_2 h_3}{h_1}\frac{\partial f}{\partial q_1}\right) + \frac{\partial}{\partial q_2}\left(\frac{h_1 h_3}{h_2}\frac{\partial f}{\partial q_2}\right) + \frac{\partial}{\partial q_3}\left(\frac{h_1 h_2}{h_3}\frac{\partial f}{\partial q_3}\right)\right]$$

In spherical polar coordinates,

$$\nabla^2 f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial f}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 f}{\partial \phi^2}$$

9. Integral Calculus with Several Independent Variables

(9장 다변수 적분)

Single-Variable Integration

For a function $f(x) = \mathrm{d}F/\mathrm{d}x$,

$$\mathrm{d}F = f(x)\mathrm{d}x$$

$$\int_{x_0}^{x_1} \mathrm{d}F = \int_{x_0}^{x_1} f(x)\,\mathrm{d}x$$

$$F(x_1) - F(x_0) = \int_{x_0}^{x_1} f(x)\,\mathrm{d}x$$

This is the integral along the x axis from x_0 to x_1

Line Integral (선적분)

Consider a differential with two independent variables:

$$\mathrm{d}F = M(x, y)\mathrm{d}x + N(x, y)\mathrm{d}y$$

For the curve C joining (x_0, y_0) and (x_1, y_1),

the line integral (path integral) is defined as

$$\int_C \mathrm{d}F = \int_C [M(x, y)\mathrm{d}x + N(x, y)\mathrm{d}y]$$

Evaluation of Line Integral

$$\int_C \mathrm{d}F = \int_C [M(x,y)\mathrm{d}x + N(x,y)\mathrm{d}y]$$

The curve can be expressed as $x = x(y)$ or $y = y(x)$, so

$$\int_C \mathrm{d}F = \int_C [M\{x,y(x)\}\mathrm{d}x + N\{x(y),y\}\mathrm{d}y]$$

$$\int_C \mathrm{d}F = \int_{x_0}^{x_1} M\{x,y(x)\}\,\mathrm{d}x + \int_{y_0}^{y_1} N\{x(y),y\}\,\mathrm{d}y$$

Example 9.1

Find the value of the line integral

$$\int_C \mathrm{d}F = \int_C [(2x+3y)\mathrm{d}x + (3x+4y)\mathrm{d}y]$$

where C is the straight-line segment given by $y = 2x + 3$ from (0,3) to (2,7).

Rewrite y in the first term in terms of x:

$$2x + 3y = 2x + 3(2x+3) = 8x + 9$$

The integral becomes

$$\int_C (2x+3y)\mathrm{d}x = \int_0^2 (8x+9)\,\mathrm{d}x = (4x^2+9x)\Big|_0^2 = 34$$

Example 9.1

Find the value of the line integral

$$\int_C \mathrm{d}F = \int_C [(2x + 3y)\mathrm{d}x + (3x + 4y)\mathrm{d}y]$$

where C is the straight-line segment given by $y = 2x + 3$ from (0,3) to (2,7).

Similarly, rewrite x in the second term in terms of y:

$$3x + 4y = 3\,(y - 3)/2 + 4y = (11y - 9)/2$$

The integral becomes

$$\int_C (3x + 4y)\mathrm{d}y = \int_3^7 \left(\frac{11}{2}y - \frac{9}{2}\right)\mathrm{d}y = \left(\frac{11}{4}y^2 - \frac{9}{2}y\right)\Bigg|_3^7 = 92$$

Hence, the total value of the line integral is 34 + 92 = 126.

Line Integrals of Exact Differentials

If $\mathrm{d}F$ is an exact differential, the line integral $\int_C \mathrm{d}F$

- depends only on the initial and final points.
- equals to F(final point) $-$ F(initial point).

For an exact differential $\mathrm{d}F$ of the function $F(x, y)$,

with the initial point (x_0, y_0) and the final point (x_1, y_1),

$$\int_C \mathrm{d}F = \int_C \left[\left(\frac{\partial F}{\partial x} \right)_y \mathrm{d}x + \left(\frac{\partial F}{\partial y} \right)_x \mathrm{d}y \right] = F(x_1, y_1) - F(x_0, y_0)$$

Example 9.2

Calculate the line integral

$$\int_C \mathrm{d}F = \int_C [(2x + 3y)\mathrm{d}x + (3x + 4y)\mathrm{d}y]$$

on the rectangular path from (0, 3) to (2, 3) and then to (2, 7).

Divide the path into C_1 from (0, 3) to (2, 3) and C_2 from (2, 3) to (2, 7)

$$\int_C \mathrm{d}F = \int_{C_1} \mathrm{d}F + \int_{C_2} \mathrm{d}F$$

On C_1, $y = 3$ (fixed) and $\mathrm{d}y = 0$, so

$$\int_{C_1} [(2x + 3y)\mathrm{d}x + (3x + 4y)\mathrm{d}y] = \int_0^2 (2x + 9)\,\mathrm{d}x = (x^2 + 9x)\Big|_0^2 = 22$$

Example 9.2

Calculate the line integral

$$\int_C \mathrm{d}F = \int_C [(2x + 3y)\mathrm{d}x + (3x + 4y)\mathrm{d}y]$$

on the rectangular path from (0, 3) to (2, 3) and then to (2, 7).

Divide the path into C_1 from (0, 3) to (2, 3) and C_2 from (2, 3) to (2, 7)

$$\int_C \mathrm{d}F = \int_{C_1} \mathrm{d}F + \int_{C_2} \mathrm{d}F = 22 + \int_{C_2} \mathrm{d}F$$

On C_2, $x = 2$ (fixed) and $\mathrm{d}x = 0$, so

$$\int_{C_2} [(2x + 3y)\mathrm{d}x + (3x + 4y)\mathrm{d}y] = \int_3^7 (6 + 4y)\,\mathrm{d}y = (6y + 2y^2)\Big|_3^7 = 104$$

The total value of the line integral is 22 + 104 = 126.

Line Integrals of Inexact Differentials

If a differential is not exact, two line integrals on curves beginning and ending at the same points will not necessarily yield the same result.

Example 9.3

Show that the differential $\mathrm{d}u = \mathrm{d}x + x\mathrm{d}y$ is inexact,
and carry out the line integral from (0, 0) to (2, 2) by two different paths:
path 1: the straight-line segment from (0, 0) to (2, 2); and
path 2: the rectangular path from (0, 0) to (2, 0) and then to (2, 2).

If a differential $M\mathrm{d}x + N\mathrm{d}y$ is exact, $(\partial M/\partial y)_x = (\partial N/\partial x)_y$

$$(\partial M/\partial y)_x = (\partial(1)/\partial y)_x = 0$$

$$(\partial N/\partial x)_y = (\partial x/\partial x)_y = 1$$

Since 0 ≠ 1, the differential is inexact.

Example 9.3

Show that the differential $\mathrm{d}u = \mathrm{d}x + x\mathrm{d}y$ is inexact,
and carry out the line integral from (0, 0) to (2, 2) by two different paths:
path 1: the straight-line segment from (0, 0) to (2, 2); and
path 2: the rectangular path from (0, 0) to (2, 0) and then to (2, 2).

Consider the line integral on path 1. The formula for path 1 is $y = x$, so

$$\int_{C_1} \mathrm{d}u = \int_{C_1} \mathrm{d}x + \int_{C_1} x\,\mathrm{d}y = \int_0^2 \mathrm{d}x + \int_0^2 y\,\mathrm{d}y$$

$$= x\Big|_{x=0}^{x=2} + \frac{y^2}{2}\Bigg|_{y=0}^{y=2} = 2 + 2 = 4$$

Example 9.3

Show that the differential $\mathrm{d}u = \mathrm{d}x + x\mathrm{d}y$ is inexact,
and carry out the line integral from (0, 0) to (2, 2) by two different paths:
path 1: the straight-line segment from (0, 0) to (2, 2); and
path 2: the rectangular path from (0, 0) to (2, 0) and then to (2, 2).

Now consider the line integral on path 2. The first segment is $y = 0$ and the second segment is $x = 2$.

$$\int_{C_2} \mathrm{d}u = \int_{C_2} \mathrm{d}x + \int_{C_2} x\,\mathrm{d}y = \int_0^2 \mathrm{d}x + \int_0^2 2\,\mathrm{d}y$$

$$= x\Big|_{x=0}^{x=2} + 2y\Big|_{y=0}^{y=2} = 2 + 4 = 6$$

Three Independent Variables

Consider

$$\mathrm{d}u = M(x, y, z)\mathrm{d}x + N(x, y, z)\mathrm{d}y + P(x, y, z)\mathrm{d}z$$

The line integral on path C is

$$\int_C \mathrm{d}u = \int_C [M(x, y, z)\mathrm{d}x + N(x, y, z)\mathrm{d}y + P(x, y, z)\mathrm{d}z]$$

Three Independent Variables

If C begins at (x_0, y_0, z_0) and ends at (x_1, y_1, z_1),

$$\int_C \mathrm{d}u = \int_C [M(x,y,z)\mathrm{d}x + N(x,y,z)\mathrm{d}y + P(x,y,z)\mathrm{d}z]$$

$$= \int_{x_0}^{x_1} M[x, y(x), z(x)]\,\mathrm{d}x + \int_{y_0}^{y_1} N[x(y), y, z(y)]\,\mathrm{d}y$$

$$+ \int_{z_0}^{z_1} P[x(z), y(z), z]\,\mathrm{d}z$$

Three Independent Variables

$$\mathrm{d}u = M(x, y, z)\mathrm{d}x + N(x, y, z)\mathrm{d}y + P(x, y, z)\mathrm{d}z$$

If $\mathrm{d}u$ is an exact differential,

$$\int_C \mathrm{d}u = \Delta u = u(x_1, y_1, z_1) - u(x_0, y_0, z_0)$$

Example 9.4

Consider the differential $\mathrm{d}u = yz\mathrm{d}x + xz\mathrm{d}y + xy\mathrm{d}z$.
Determine whether the differential is exact, and evaluate the line integral

$$\int_C \mathrm{d}u = \int_C (yz\mathrm{d}x + xz\mathrm{d}y + xy\mathrm{d}z)$$

on the path from (0, 0, 0) to (3, 3, 3) corresponding to $x = y = z$.

For $\mathrm{d}u = M\mathrm{d}x + N\mathrm{d}y + P\mathrm{d}z$, the exactness test is

$$\left(\frac{\partial M}{\partial y}\right)_{x,z} = \left(\frac{\partial N}{\partial x}\right)_{y,z}, \qquad \left(\frac{\partial N}{\partial z}\right)_{x,y} = \left(\frac{\partial P}{\partial y}\right)_{x,z}, \qquad \left(\frac{\partial M}{\partial z}\right)_{x,y} = \left(\frac{\partial P}{\partial x}\right)_{y,z}$$

and for the given differential,

$$z = z, \qquad x = x, \qquad y = y$$

so the differential is exact.

Example 9.4

Consider the differential $\mathrm{d}u = yz\mathrm{d}x + xz\mathrm{d}y + xy\mathrm{d}z$.
Determine whether the differential is exact, and evaluate the line integral

$$\int_C \mathrm{d}u = \int_C (yz\mathrm{d}x + xz\mathrm{d}y + xy\mathrm{d}z)$$

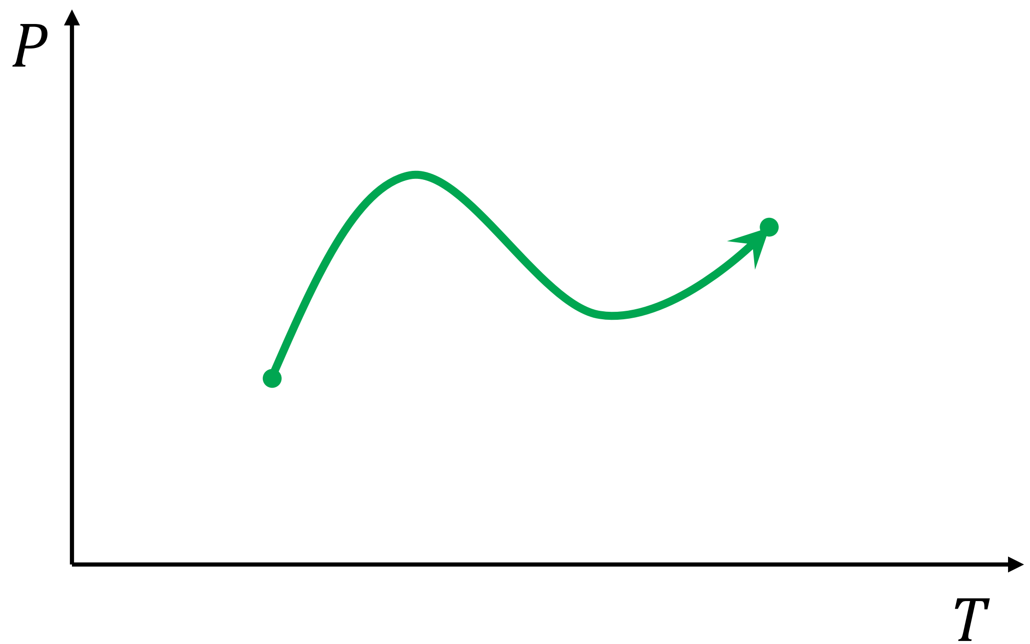
on the path from (0, 0, 0) to (3, 3, 3) corresponding to $x = y = z$.

Since the differential is exact and $u = xyz$,

$$\int_C \mathrm{d}u = u(3,3,3) - u(0,0,0) = 3 \times 3 \times 3 - 0 = 27$$

Line Integrals in Thermodynamics

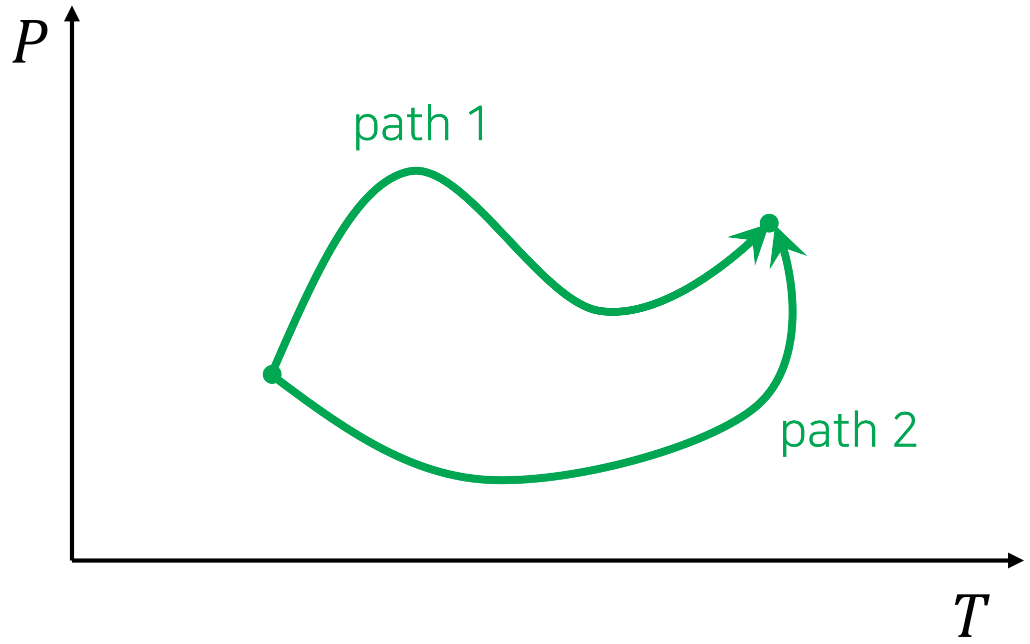
- Equilibrium state: a point in a space
- Reversible process: a line integral



Not in Mortimer

Path Dependence

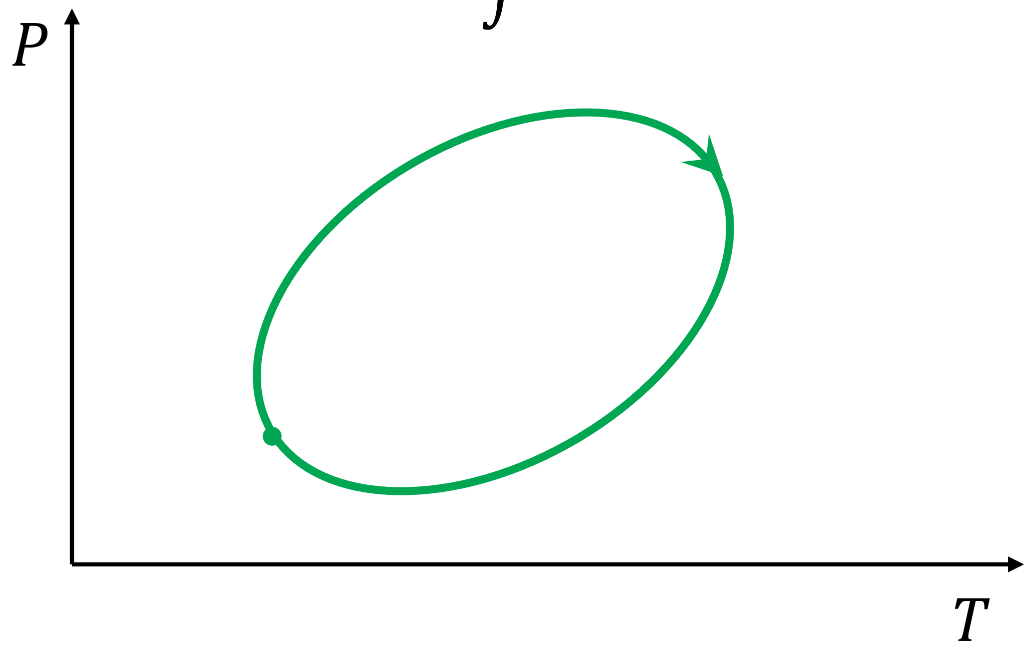
- Exact differential: path-independent integral
- Inexact differential: path-dependent integral



Cyclic Process (순환과정)

Process that begins and ends at the same state

$$\oint du$$



Cyclic Process (순환과정)

- If $\mathrm{d}u$ is exact,

$$\oint \mathrm{d}u = 0$$

  - *e.g.* $\oint \mathrm{d}U = 0$ always

- If $\mathrm{d}u$ is inexact, it is not necessary
  - *e.g.* $\oint \delta w \neq 0$ for some processes (see Example 9.5)